

A Small-Cardinal Partial Answer for α -Large Families

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2026-06-23

Status. Partial result. This note records a direct restriction consequence of the construction of Argyros–Motakis [1]. It answers their α -large-family question for all infinite cardinals at most the continuum.

Source question. Argyros–Motakis ask the following question on page 14 of [1], after recalling the Lopez-Abad–Todorcevic characterization of the ordinary large-family case:

Let κ be an infinite cardinal number which is not ω -Erdos and α be a countable ordinal number. Does there exist an α -large, hereditary and compact family of finite subsets of κ ?

Definitions used. If A is an infinite set and $\mathcal{B} \subseteq [A]^{<\omega}$, then $\mathcal{B}$ is α -large if for every infinite $B \subseteq A$ there is a one-to-one map $\phi : \mathbb{N} \rightarrow B$ such that $\phi(F) \in \mathcal{B}$ for every Schreier set $F \in \mathcal{S}_\alpha$. The family is compact when it is closed in the product space 2^A and all of its elements are finite. It is hereditary when $G \subseteq F \in \mathcal{B}$ implies $G \in \mathcal{B}$.

Proof Intuition. Argyros–Motakis already construct, for each countable α , an α -large hereditary compact family $\mathcal{G}_\alpha$ on the Cantor set $2^\mathbb{N}$, which has cardinality $2^{\aleph_0}$. Nothing in the definition of α -largeness requires using all Cantor branches. If $A \subseteq 2^\mathbb{N}$, then every infinite subset of A is still an infinite subset of the Cantor set, so the same witness sequence supplied by $\mathcal{G}_\alpha$ stays inside A . Compactness also restricts because the coordinate-extension map $2^A \rightarrow 2^{2^\mathbb{N}}$ is a topological embedding.

Theorem 1. *Let κ be an infinite cardinal with $\kappa \leq 2^{\aleph_0}$, and let $\alpha < \omega_1$. Then there exists an α -large, hereditary and compact family of finite subsets of κ .*

Proof. Choose a set $A \subseteq 2^\mathbb{N}$ of cardinality κ , and identify κ with A . By Theorem 2.20 of Argyros–Motakis [1], there is an α -large, hereditary and compact family $\mathcal{G}_\alpha \subseteq [2^\mathbb{N}]^{<\omega}$. Define

$$\mathcal{H}_\alpha(A) = \{F \in \mathcal{G}_\alpha : F \subseteq A\}.$$

Clearly $\mathcal{H}_\alpha(A) \subseteq [A]^{<\omega}$, and heredity follows immediately from the heredity of $\mathcal{G}_\alpha$.

We next check compactness. Let $j : 2^A \rightarrow 2^{2^\mathbb{N}}$ be the map that extends a subset of A by zero outside A . This is a topological embedding onto the closed set $\{x : x(\tau) = 0 \text{ for all } \tau \notin A\}$. If x belongs to the closure of $\mathcal{H}_\alpha(A)$ in 2^A , then $j(x)$ belongs to the closure of $\mathcal{G}_\alpha$ in $2^{2^\mathbb{N}}$. Since $\mathcal{G}_\alpha$ is compact, $j(x) \in \mathcal{G}_\alpha$. In particular $j(x)$ is a finite subset of $2^\mathbb{N}$, supported inside A , so x is a finite subset of A and $x \in \mathcal{H}_\alpha(A)$. Hence $\mathcal{H}_\alpha(A)$ is closed in 2^A , and therefore compact.

Finally let $B \subseteq A$ be infinite. Since B is also an infinite subset of $2^\mathbb{N}$, the α -largeness of $\mathcal{G}_\alpha$ gives a one-to-one map $\phi : \mathbb{N} \rightarrow B$ such that $\phi(F) \in \mathcal{G}_\alpha$ for every $F \in \mathcal{S}_\alpha$. But $\phi(F) \subseteq B \subseteq A$, hence $\phi(F) \in \mathcal{H}_\alpha(A)$. Thus $\mathcal{H}_\alpha(A)$ is α -large on A . $\square$

Remark 1. *This proves the Argyros–Motakis question for all infinite $\kappa \leq 2^{\aleph_0}$. The proof gives no information about non- ω -Erdos cardinals strictly above the continuum. In particular, the full question remains open from this argument.*

Verification and novelty check. The lightweight run indexes were searched for arXiv:1302.0715 and the core phrases “alpha-large”, “omega-Erdos”, and “hereditary compact family”, with no existing packet or attempt found. A bounded web search for the exact question and close variants found the source paper but no later full answer. The result above is only the restriction subcase $\kappa \leq 2^{\aleph_0}$, not a solution of the full problem.

References

- [1] S. A. Argyros and P. Motakis, *α -Large Families and Applications to Banach Space Theory*, arXiv:1302.0715.